

Automated Cross-Site Backup & Restore System

VPS → Proxmox VE (WireGuard + rsync + Hardlinks)

Author: Jacek Kapłon

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1 Architecture Overview



Figure 1: Automated Cross-Site Backup Architecture

This document describes a fully reproducible secure backup pipeline between a remote VPS and a Proxmox VE host using WireGuard VPN, rsync-based incremental snapshots and automated retention policy.

2 WireGuard Configuration

2.1 Install WireGuard

On both VPS and Proxmox:

```
apt update
apt install wireguard -y
```

2.2 Generate Keys (Both Sides)

```
wg genkey | tee /etc/wireguard/private.key | \
wg pubkey > /etc/wireguard/public.key
chmod 600 /etc/wireguard/private.key
```

2.3 Proxmox Configuration

File: /etc/wireguard/wg-bkp-pve.conf

```
[Interface]
Address = 10.50.50.1/24
ListenPort = 51855
PrivateKey = <PVE_PRIVATE_KEY>

[Peer]
```

```
PublicKey = <VPS_PUBLIC_KEY>
AllowedIPs = 10.50.50.2/32
```

2.4 VPS Configuration

File: /etc/wireguard/wg-backup-vps.conf

```
[Interface]
Address = 10.50.50.2/24
PrivateKey = <VPS_PRIVATE_KEY>

[Peer]
PublicKey = <PVE_PUBLIC_KEY>
Endpoint = <PVE_PUBLIC_IP>:51855
AllowedIPs = 10.50.50.1/32
PersistentKeepalive = 25
```

2.5 Start Tunnel

```
wg-quick up wg-bkp-pve
wg-quick up wg-backup-vps
```

Test:

```
ping 10.50.50.1
ping 10.50.50.2
```

3 Backup User Setup on Proxmox

```
useradd -m -s /bin/bash vpsbackup
passwd -d vpsbackup

mkdir -p /mnt/usb-backup/vps-backups
chown -R vpsbackup:vpsbackup /mnt/usb-backup
```

4 SSH Key Authentication

On VPS:

```
ssh-keygen -t ed25519
ssh-copy-id vpsbackup@10.50.50.1
```

Optional hardening on Proxmox:

Edit /etc/ssh/sshd_config

```
PasswordAuthentication no
PermitRootLogin no
```

Restart SSH:

```
systemctl restart ssh
```

5 Initial Full Backup Script

File: /root/vps-backup-initial.sh

```
#!/bin/bash

DATE=$(date +%F)
REMOTE_USER="vpsbackup"
REMOTE_HOST="10.50.50.1"
REMOTE_BASE="/mnt/usb-backup/vps-backups"
REMOTE_TARGET="$REMOTE_BASE/$DATE"

echo "[1/3] Creating directory..."
ssh $REMOTE_USER@$REMOTE_HOST "mkdir -p $REMOTE_TARGET"

echo "[2/3] Running full rsync..."
rsync -aAX \
--delete \
--exclude={"/proc/*","/sys/*","/dev/*","/tmp/*","/run/*","/mnt/*","/media/*","/lost+found"} \
/ \
$REMOTE_USER@$REMOTE_HOST:$REMOTE_TARGET

echo "[3/3] Updating latest symlink..."
ssh $REMOTE_USER@$REMOTE_HOST \
"ln -sf $REMOTE_TARGET $REMOTE_BASE/latest"

echo "Initial FULL backup completed."
```

Make executable:

```
chmod +x /root/vps-backup-initial.sh
```

6 Incremental Backup Script

File: /root/vps-backup-incremental.sh

```
#!/bin/bash

DATE=$(date +%F)
REMOTE_USER="vpsbackup"
REMOTE_HOST="10.50.50.1"
REMOTE_BASE="/mnt/usb-backup/vps-backups"
REMOTE_TARGET="$REMOTE_BASE/$DATE"
REMOTE_LATEST="$REMOTE_BASE/latest"

echo "[1/4] Creating directory..."
ssh $REMOTE_USER@$REMOTE_HOST "mkdir -p $REMOTE_TARGET"

echo "[2/4] Running incremental rsync..."
rsync -aAX \
--delete \
--link-dest="$REMOTE_LATEST" \
--exclude={"/proc/*","/sys/*","/dev/*","/tmp/*","/run/*","/mnt/*","/media/*","/lost+found"} \
/ \
$REMOTE_USER@$REMOTE_HOST:$REMOTE_TARGET
```

```
echo "[3/4] Updating latest symlink..."
ssh $REMOTE_USER@$REMOTE_HOST \
"ln -sf $REMOTE_TARGET $REMOTE_LATEST"

echo "[4/4] Incremental backup complete."
```

7 Automation

On VPS:

```
crontab -e
```

Add:

```
0 2 * * * /root/vps-backup-incremental.sh >> \
/var/log/vps-backup.log 2>&1
```

On Proxmox (Retention 14 days):

```
30 3 * * * find /mnt/usb-backup/vps-backups \
-maxdepth 1 -type d -mtime +14 -exec rm -rf {} \;
```

8 Restore Workflow

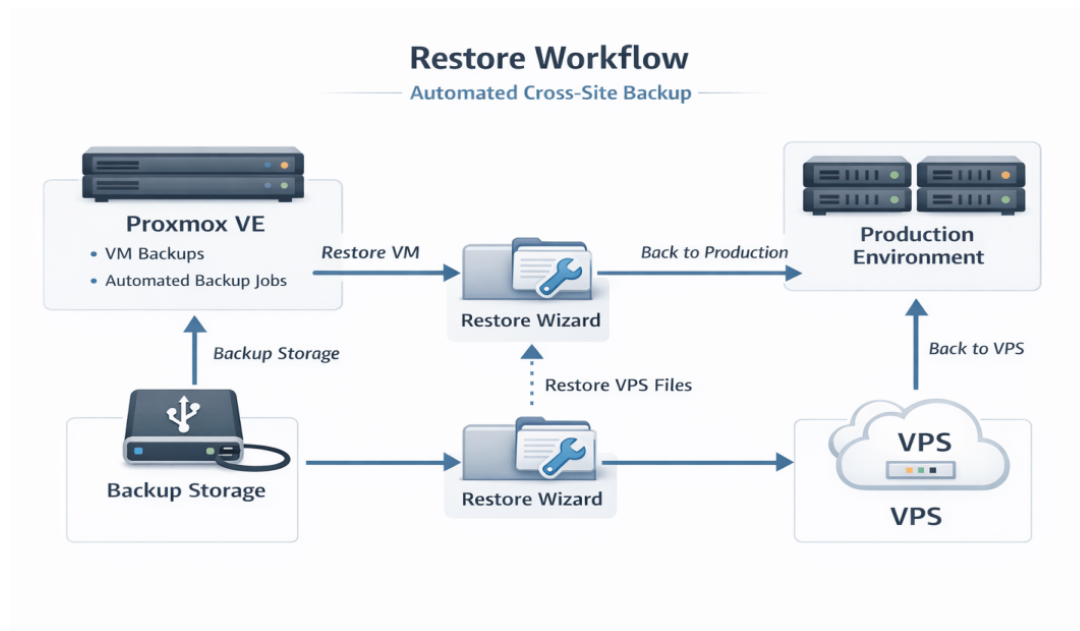


Figure 2: Restore Workflow

8.1 Download Backup

```
scp -r \
vpsbackup@10.50.50.1:/mnt/usb-backup/vps-backups/latest \
/root/restore/
```

8.2 Restore Files

```
rsync -aAXv /root/restore/latest/ /
```

8.3 Restart Services

```
systemctl daemon-reload  
systemctl restart <service_name>
```

9 Challenges & Lessons Learned

9.1 WireGuard Debugging

Misconfigured AllowedIPs or firewall rules can silently break connectivity. Always verify routing and peer state using `wg show` and `ip route`.

9.2 Security First

SSH key authentication reduces attack surface. Never expose password login on public interfaces.

9.3 Snapshot Efficiency

Hardlink-based incremental backups provide enterprise-level storage optimization without dedicated backup software.

9.4 Test Restores

Backup without restore validation is not disaster recovery. Always test recovery procedures.

10 Technical Summary

This system provides:

- Encrypted VPN transport (WireGuard)
- SSH key-based secure authentication
- Automated daily incremental backups
- Hardlink-based storage efficiency
- 14-day retention policy
- Full disaster recovery workflow